Q.1 What is the best case time complexity of merge sort?

A. O((n^2logn)
B. O((n^2)
C. O((nlogn)
D. O((nlogn^2)

Q.2 The important factors to measure the running time of the brute-force 2-d maxima algorithm are

A. Count the number of steps of the pseudo code that are executed
B. The number of comparisons that are performed
C. All of above
D. Count the number of times an element of P is accessed

Q.3 In 3-Dimensional space, a point P has _____ coordinate(s).

A. (0, Y)
B. (X, 0)
C. (X, Y, Z)
D. (X, Y)

Q.4 Plane-Sweep algorithm can run in _____ time.

A. Θ(n log)
B. Θ(n*n)
C. Θ(n)
D. Θ(n log n)

Q.5 The _____ is the weighted sum of running times with weights.

A. worse-case time
B. both best-case and average-case.
C. best-case time
D. average-case time

Q.6 An algorithm is a mathematical entity, which is independent of _____.

A. Compiler and Programming language
B. Programming language, Compiler and Machine
C. Machine and Programming language
D. Programming language

Q.7 There are no hard formal rules to the syntax of the _____ code.

A. Basic

B. Programming

C. Pseudo

D. Assembly

Q.8 In the following loop statement: for i ← 1 to n For loop executes _____ times.

A. n

B. It is not correct statement

C. log n

D. n2

Q.9 The definition of theta-notation relies on proving _____ asymptotic bound.

A. upper

B. lower

C. both lower and upper

D. one

Q.10 _____ time is the maximum running time over all legal inputs.

A. Best-case

B. Good-case

C. Worst-case

D. Average-case

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